

Computer Vision - 2026

Lecture #02. Convolutional Neural Networks

Lectures by Alexei Kornaev ^{1,2}

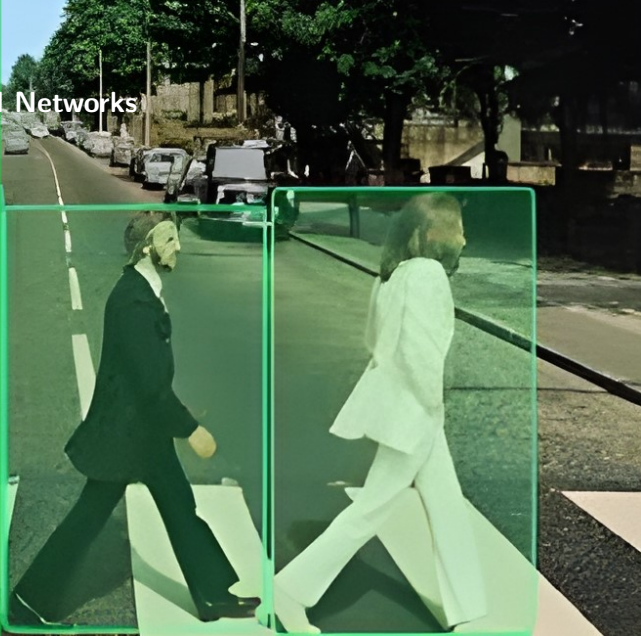
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January 26, 2026



Agenda

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

- ① Outcomes
- ② Key Ideas in CNNs
- ③ Tips and Tricks in CNNs
- ④ CNNs: the Current State and Prospects
- ⑤ Conclusion

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Section 1. Outcomes

Outcomes

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

This week's lecture and practice on CNNs is designed to provide a deep understanding of the theoretical concepts and practical skills necessary for working with CNN-based architectures. By the end of this lecture, students will be able to:

- 1 Understand core concepts: convolutions, filters, feature maps, padding, pooling, and key CNN architectures (e.g., ResNet, DenseNet, ConvNeXt).
- 2 Apply transfer learning and fine-tuning to adapt pre-trained models for new tasks using tools like PyTorch.
- 3 Implement CNNs on datasets like MNIST and CIFAR-10, bridging the gap from educational tasks to real-world applications.

Key Takeaway: Transfer learning accelerates solving practical problems by leveraging pre-trained models.

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Section 2. Key Ideas in CNNs

Physiology of cats

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

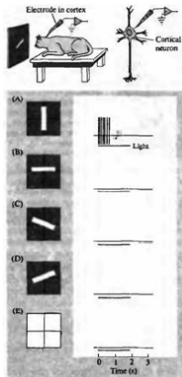


Figure: Responses of the cat's visual cortex cell.

Implementation in math.

The horizontal derivative kernel approximates $\frac{\partial I}{\partial x}$ using **finite differences**. For images, this translates to computing intensity changes along the x-axis. The Sobel kernel for horizontal derivative is:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

This kernel computes:

$$\frac{\partial I}{\partial x} \approx (I(x+1, y) - I(x-1, y))$$

and incorporates smoothing to reduce noise. It detects horizontal edges by highlighting intensity changes from left to right.

Hands-on coding: **Kernels** [CV-2026]

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

#	Filter	Description
1	Median Filtering	Replaces each pixel with the median value of its neighborhood.
2	Gaussian Blur	A Gaussian kernel. Reduces noise but blurs edges.
3	Bilateral Filtering	Smooths the image while preserving edges by considering both spatial and intensity differences.
4	Non-Local Means Denoising	Removes noise by averaging similar patches across the image. Computationally intensive.
5	Histogram Equalization	Redistributes pixel intensities to improve contrast. Simple and effective for enhancing details.
6	Edge Detection (Canny)	Detects edges using gradient-based methods and hysteresis thresholding. Preserves strong edges.
7	Thresholding	Converts a grayscale image to binary based on a threshold value. Simple pixel-wise operation.
8	Morphological Operations (Closing)	Closes small defects in binary images using a structuring element.
9	Inpainting	Fills missing or corrupted regions in the image using surrounding information. Solves partial differential equations.

Table: Filters.

Padding, Pooling, and Striding

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

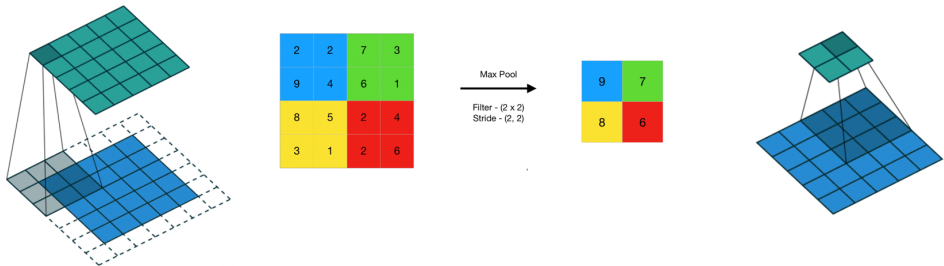


Figure: Padding (left), Pooling (middle), and Striding (right). CNNs by Neurohive.

2D Convolution Applied to an Image

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

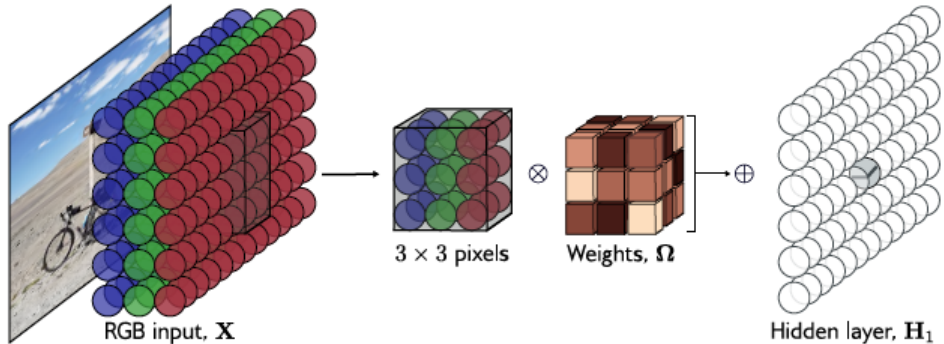


Figure: The image is treated as a 2D input with three channels corresponding to the red, green, and blue components. With a 3×3 kernel, each pre-activation in the first hidden layer is computed by pointwise multiplying the $3 \times 3 \times 3$ kernel weights with the $3 \times 3 \times 3$ RGB image patch centered at the same position, summing, and adding the bias. To calculate all the pre-activations in the hidden layer, we “slide” the kernel over the image in both horizontal and vertical directions. The output is a 2D layer of hidden units. To create multiple output channels, we would repeat this process with multiple kernels, resulting in a 3D tensor of hidden units at hidden layer $\mathbf{H}_1$ [Prince, 2023]

Result of a convolution

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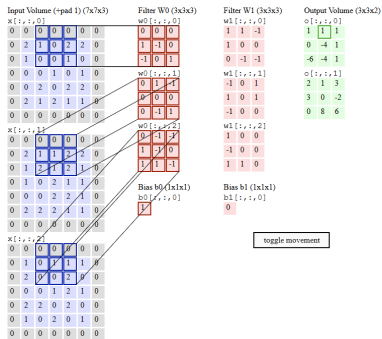
Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion



The result of convolving an image

of size $[q, q]$ with a kernel of size $[k, k]$, with padding $[p, p]$ and stride $[s, s]$, is a matrix of size $[r, r]$:

$$r = \frac{q - k + 2p}{s} + 1 = \frac{5 - 3 + 2}{2} + 1 = 3.$$

Figure: CS231n: CNNs.

Hands-on the book by Howard and Gugger [2020]

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CNN utilizes the features of the visual cortex, where simple cells are activated by simple features (such as lines), and complex cells by combinations of activations of simple cells. The CNN is associated with the mathematical operation of convolution for reducing matrix sizes.

Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

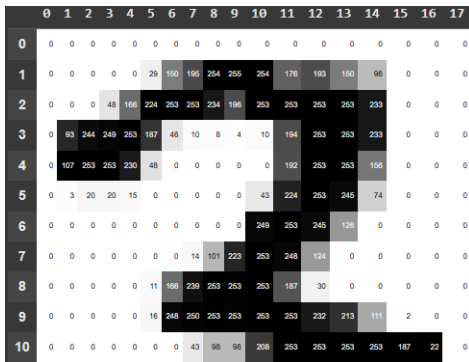


Figure: Check the code: 13 convolutions.ipynb, and 14 resnet.ipynb

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Section 3. Tips and Tricks in CNNs

Residual Connections (ResNet idea)

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Problem: very deep CNNs suffer from degradation:

- training error increases with depth
- gradients vanish/explode

Residual block formulation: $y = F(x, W) + x$

where:

- x is the input feature map
- $\mathcal{F}$ is the residual function ($\text{Conv} \rightarrow \text{BN} \rightarrow \text{ReLU}$)

Key intuition:

- learn **corrections**, not full mappings
- identity mapping is easy to preserve

Why Residual Connections Work

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Optimization perspective:

- residual path creates a short gradient route
- improves conditioning of the loss landscape

Gradient flow:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \left(\frac{\partial \mathcal{F}}{\partial \mathbf{x}} + I \right)$$

Practical consequence:

- deeper networks train reliably
- ResNet-50 often outperforms plain CNN-20

Batch Normalization (BN)

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Goal: stabilize and accelerate training

For activations x in a mini-batch:

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta$$

Effects:

- reduces internal covariate shift
- allows higher learning rates
- acts as weak regularizer

Typical order:

Conv $\rightarrow$ BN $\rightarrow$ ReLU

BatchNorm vs LayerNorm (CNN context)

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

	BatchNorm	LayerNorm
Normalization axis	batch	channels
Batch-size dependent	yes	no
CNN default	yes	no
Transformer default	no	yes
Small batch stability	weak	strong

Rule of thumb:

- CNNs $\Rightarrow$ BatchNorm
- ViTs / small batches $\Rightarrow$ LayerNorm

Transfer Learning in CNNs

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Idea: reuse representations learned on large datasets (e.g. ImageNet)

Why it works:

- early layers learn generic features (edges, textures)
- later layers are task-specific

Two standard strategies:

- ① **Feature extraction:** freeze backbone, train head
- ② **Fine-tuning:** unfreeze top- k layers and retrain

Fine-Tuning: Practical Protocol

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Recommended steps:

- 1 Train classifier head with frozen backbone
- 2 Unfreeze last block(s)
- 3 Reduce learning rate by $10\times$
- 4 Monitor validation loss carefully

Common mistakes:

- unfreezing too early
- using same LR for all layers
- ignoring catastrophic forgetting

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Section 4. CNNs: the Current State and Prospects

CNN Architectures: Comparison

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Model	Year	Key idea	Params	Use case
AlexNet	2012	Deep conv + ReLU	~60M	History
VGG	2014	Small 3×3 convs	~138M	Baseline
ResNet	2015	Residual connections	25-100M	Default CNN
MobileNet	2017	Depthwise conv	~4M	Edge devices
EfficientNet	2019	Compound scaling	5-66M	Accuracy/efficiency
ConvNeXt	2022	Modernized CNN	28-90M	CNN vs ViT

Observation: modern CNNs adopt transformer-style design choices.

CNN vs ViT: When to Use What

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

	CNN	ViT
Small datasets	✓	
Inductive bias	strong	weak
Training stability	high	lower
Scalability	medium	high
Open-vocabulary	limited	strong

Rule of thumb:

- limited data / edge $\Rightarrow$ CNN
- large data / multimodal $\Rightarrow$ ViT

Paper Reading

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

The following review papers on CNNs [[Kiranyaz et al., 2021](#); [Li et al., 2021](#)] are the most cited in 2025:

- ① 1D Convolutional Neural Networks and Applications - A Survey
- ② A Survey of Convolutional Neural Networks: Analysis, Applications, and Prospects

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Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

Section 5. Conclusion

Conclusion

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Summary:

Outcomes

Key Ideas in
CNNs

Tips and
Tricks in
CNNs

CNNs: the
Current State
and Prospects

Conclusion

- Convolutions reduce parameters and improve efficiency by enforcing structured connectivity, enabling deeper models with less overfitting.
- Thoughtful architecture design (e.g., convolutions) outperforms theoretical but impractical fully connected networks in real-world applications.
- Residual connections, batch normalization, and transfer learning form the practical backbone of modern CNN training.

Bibliography

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Outcomes

Key Ideas in CNNs

Tips and Tricks in CNNs

CNNs: the Current State and Prospects

Conclusion

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